BAS-202- Mathematics-II

L T P C
3 1 0 4

MODULE-I

Numerical Techniques – I

Zeroes of transcendental and polynomial equation using Bisection method, Regula-falsi method and Newton-Raphson method, Rate of convergence of above methods. Interpolation: Finite differences, difference tables, Newton‟s forward and backward interpolation , Lagrange‟s and Newton‟s divided difference formula for unequal intervals.

Numerical Techniques –II

Solution of system of linear equations, Gauss- Seidal method, Crout method. Numerical differentiation, Numerical integration , Trapezoidal , Simpson‟s one third and three-eight rules, Solution of ordinary differential (first order, second order and simultaneous) equations by Euler‟s, Picard‟s and forth-order Runge-Kutta mehthods.

MODULE-II

Statistical Techniques - I

Moments, Moment generating functions, Skewness, Kurtosis, Curve fitting, Method of least squares, Fitting of straight lines, Polynomials, Exponential curves etc., Correlation, Linear, non –linear and multiple regression analysis, Probability theory.

Statistical Techniques - II

Binomial, Poisson and Normal distributions, Sampling theory (small and large), Tests of significations, Chi-square test, t-test.

MODULE-III

Multiple Integrals

Double and triple integral, Change of order, Change of variables, Beta and Gamma functions, Application to area, volume, Dirichlet integral and applications.

Vector Calculus

Line, surface and volume integrals, Statement and problems of Green‟s, Stoke‟s and Gauss divergence theorems (without proof).

Text Books:-

1. H.K.Dass, Higher Engineering Mathmatics, S.Chand Publications.
2. B.S.Grewal, Engineering Mathematics, Khanna Publishers,2004.

Reference Books:-

1. R.K.Jain & S.R.K.Iyenger, Advance Engineering Mathematics, Narosa Publishing House, 2002.
2. B.S.Grewal, Higher Engineering Mathematics, Khanna Publishers,2005.
3. E.Kreyszig, Advanced Engineering Mathematics, John Wiley & Sons,2005.
4. C.Ray Wylie & Louis C. Barrett, Advanced Engineering Mathematics, Tata Mc Graw-Hill Publishing Company Ltd.2003
5. Peter V. O‟Neil, Advanced Engineering Mathematics, Thomson (Cengage) Learning,2007.